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# Agroforestry system reduces subsurface lateral flow and nitrate loss in Jiangxi Province, China

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ABSTRACT

Subsurface lateral flow is an important pathway responsible for agricultural non-point source pollution and may be affected by land use. An agroforestry system, citrus (*Citrus reticulata*) tree intercropped with peanut (*Arachis hypogaea*) crop and a mono peanut cropping system were compared over the period from 2003 to 2005 in Jiangxi Province, China. The objectives of this study were (i) to identify subsurface lateral flow by monitoring soil matric potential and $NO_3$-N concentration in soil water, and (ii) to estimate subsurface lateral flow and associated $NO_3$-N loss by modelling the water budget using Hydrus-2D. The dynamics of soil water, either during a particular storm or on an annual basis, demonstrated that subsurface lateral flow generated along the slopes under the two systems. The agroforestry system had a larger domain and a longer resident time of water saturation in the deeper soil layers than the mono cropping system, suggesting that the agroforestry system may have increased water retention capacity of subsurface soil due to its deeper root system. The simulated annual water budget showed that, compared with the mono cropping system, the agroforestry system reduced subsurface lateral flow by 9.2% of annual rainfall, which was equivalent to the amount of precipitation predicted to be reduced by interception. The two cropping systems received the similar amount of organic and inorganic N fertilizers (160.0–170.0 kg N $ha^{-1}$ $a^{-1}$), the total amount of N inputs in agroforestry system was smaller as it may have received less biological fixed N by about 0 to 70 kg N $ha^{-1}$ $a^{-1}$ due to the smaller effective area occupied by peanut crops. $NO_3$-N concentration measured in soil water in the agroforestry system was low within the soil profile (0.6–7.6 mg $L^{-1}$) and had little seasonal variation. However, in the mono cropping system the $NO_3$-N concentration had two peaks in a year, which ranged from 14 to 52 mg $L^{-1}$ at all soil depths between 0.20 and 0.85 m, and was higher on the lower slope than on the upper slope position. The estimated $NO_3$-N loss associated with subsurface lateral flow ranged from 45 to 64 kg $ha^{-1}$ $a^{-1}$ in the mono cropping system and from 16 to 48 kg $ha^{-1}$ $a^{-1}$ in the agroforestry system. The smaller $NO_3$-N loss in the agroforestry system was probably attributed to the smaller total N inputs and/or to the reduced subsurface lateral flow. However, the underlying mechanisms and the effectiveness of agroforestry systems need further study.



## 1. Introduction

Intensive agriculture increases the environmental risks to surface and ground water quality (Groffman, 2000; Jalali, 2005). Overland flow and subsurface lateral flow are two dominant hydrological pathways for the transport of sediments, nutrients and pollutants (Sims et al., 1998). Overland flow has been intensively studied, as it can be easily monitored and is considered to be responsible for a large proportion of nutrient and pollutant loss from farmland (Edwards and Owens, 1991). Gravity induces subsurface lateral flow along slopes when sufficient vertical flow is impeded, and a saturated zone is formed over a less permeable layer within a soil profile (Shaw et al., 2001) - in extreme cases over bedrock (Onda et al., 2004). Subsurface lateral flow is often considered to be a slow process in natural ecosystems and it leads to spatial variations in soluble nutrients and pollutants (Fitzpatrick et al., 1996; Sommer and Stahr, 1996). However, subsurface lateral flow in agricultural catchments is not well understood.

Soil compaction (Horn and Smucker, 2005) and the intensified eluviation of clay due to improper soil management (Sommer

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and Stahr, 1996; Oygarden et al., 1997) may result in a reduced hydraulic conductivity of agricultural soils at lower depths. The creation of a perched water table over less permeable soil layers can be a relatively quick process (Wilson et al., 1990; Latron and Gallart, 2008), resulting in agricultural catchments, in a rapid subsurface lateral flow along slopes (McDaniel et al., 2007; Tang, 2005). Although it is still a challenge to monitor and quantify subsurface lateral flow, it has been identified from soil moisture dynamics (Giles, 2005; Lin, 2006; Lin and Zhou, 2008) and with stream hydrochemographs (Liu et al., 2004; Inamdar and Mitchell, 2007). Although more attention has recently been paid to identify the effects of land use, landform and soil management (Honisch et al., 2002; Ridolfi et al., 2003; Bechtold et al., 2007), few studies have attempted to quantify these effects by applying two-dimensional modelling.

Agroforestry systems involve growing woody herbaceous species and perennials in association with food crops on the same piece of land (David and Raussen, 2003). These systems are increasingly being considered as a means of addressing environmental problems because they can reduce soil and nutrient loss through overland flow. Udawatta et al. (2002) report that an agroforestry watershed consisting of trees plus pasture in northeastern Missouri reduced $NO_3$-N and P loss by 37% and 17%, respectively, compared with a mono corn-cropping watershed, owing to the reduced surface runoff. Recently, Allen et al. (2006) and Nair et al. (2007) observed that nutrient concentrations in soil were lower in the agroforestry system consisting of trees and crops than in paired crops and pasture, suggesting that the deeper-rooted agroforestry system may have had enhanced nutrient uptake and so reduced the risks to the environment. Ridolfi et al. (2003) reported that deep-rooted tree systems influenced the spatial variation of soil moisture. This suggests that agroforestry systems may influence nutrient loss through subsurface lateral flow.

We hypothesized that deep-rooted agroforestry would reduce subsurface lateral flow and $NO_3$-N loss. An agroforestry system consisting of citrus (*Citrus reticulata*) trees and peanut (*Arachis hypogaea*) crops plus a mono peanut cropping system were investigated from 2003 to 2005. The objectives of this study were (i) to identify subsurface lateral flow by monitoring soil matric potential and $NO_3$-N concentration in soil water within soil profiles along a slope and (ii) to quantify subsurface lateral flow and associated $NO_3$-N loss using Hydrus-2D. The underlying mechanisms were also explored.

## 2. Materials and methods

### 2.1. Experimental site

The research area is representative of a typical moist, subtropical climate. It has a mean annual temperature of 17.7 °C, a maximum daily temperature of around 40 °C in summer, an annual average of 262 frost-free days, an annual rainfall of 1786 mm and an annual potential evaporation of 1229.1 mm. About 50% of rainfall occurs during the rainy season from March to early July and about 50% of evaporation occurs during the dry season from middle July to November. The experimental site is located in a small agricultural catchment, about 5 km from the Ecological Experimental Station of Red Soil, the Chinese Academy of Sciences, in Yingtan, Jiangxi Province, China (28°15′N, 116°55′E). Detailed information on the study catchment can be found in Tang et al. (2008) and Zepp et al. (2005). Briefly, the elevation at the catchment is from 41 to 55 m above sea level. The geology in the region consist of a weakly weathered Cretaceous sandstone underlying a deeply weathered Quaternary red clay, resulting in the formation of lateritic profiles on the hills: surficial clayey, sandy or their mixture deposits, ferruginized caprock and mottled zone, overlying weakly weathered sandstone. The depth of the soils overlying the impermeable sandstone ranges from 1.8 to 2.0 m. The soil developed from Quaternary clay is classified as a loam clay Ultisol by the Soil Taxonomy (Soil Survey Staff, 2010).

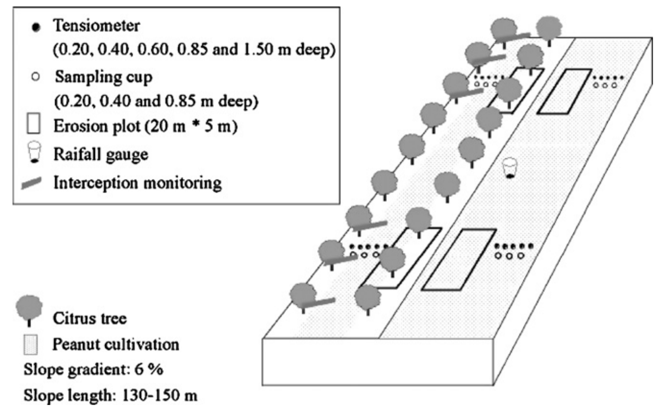


**Fig. 1.** Sketch of the instrumentation at different slope positions in the agroforestry system consisting of citrus trees and peanut crop and the mono peanut cropping system.

Two adjacent cropping systems were compared. These were a mono cropping system with peanut (*Arachis hypogaea*) and an agroforestry system consisting of citrus (*Citrus reticulata*) and an intercropped peanut crop within the alleys (Fig. 1). The slope gradient was about 6% and the slope length about 130–150 m. In 1989 citrus trees were transplanted in rows along the slope after tea (*Camellia sinensis*) plants were cleared, and the slope was converted into agroforestry and cropland. In 1999, citrus trees were partially replaced with peanut crop because of a deterioration in the orange market. The citrus trees were spaced in 4 m squares along the slope and were about 3 m high in 2003. The citrus root system was distributed to a depth of 1.10 m and the diameter of the tree crown was about 3.0–4.0 m, as measured in 2003 (Jing, 2004). The phenology, fertilization and soil management of the peanut crop were kept the same in both cropping systems. The peanut crop was sown in the middle of April along the slope at a spacing of 0.20 m × 0.30 m and harvested in the middle of August. The maximum leaf area index (LAI) of the peanut crop was 3.2, as measured in late May (Jing, 2004). In the agroforestry system, the peanut crop was planted 1.5 m away from the tree lines to avoid shading effects from the citrus canopy.

Urea was applied to the peanut crop twice a year, once as a basal fertilizer at a rate of 115.0 kg N $ha^{-1}$ on sowing, and again as a flowering fertilizer at a rate of 46 kg N $ha^{-1}$ in mid May. In the agroforestry system, additional urea and pig manure were applied to the citrus trees twice a year, once as a rejuvenation fertilizer in December, one month after the fruit harvest, and another as a blossoming fertilizer in the middle of March, one month before blossoming. The annual amount of urea applied to the citrus trees came to 100 kg N $ha^{-1}$ $a^{-1}$ and the annual amount of manure was equivalent to 19.0–22.0 kg N $ha^{-1}$ $a^{-1}$. The total N applied was, then, different in the two land uses, being about 160.0–170.0 kg N $ha^{-1}$ $a^{-1}$ in the agroforestry system, and 161.0 N $ha^{-1}$ $a^{-1}$ in the mono cropping system.

### 2.2. Field monitoring and sampling

The fields were equipped to monitor rainfall and its interception, overland flow, and soil matric potential profiles on different slope positions during the period from 2003 to 2005 (Fig. 1). Precipitation was measured using a rain gauge (Nanjing Hydrologic Instrument Company, China) placed in the mono cropping system. The rain gauge was connected to a data logger (DL2e, Delta T Inc. UK) and the data were recorded automatically at 10-minute intervals. Through-fall was monitored to calculate precipitation interception in the agroforestry system using half-open polyvinyl chloride (PVC) tubes with three replicates on both the upper and lower slope positions. The PVC tubes, 2 m long and 0.20 m in diameter, were placed in a lateral direction underneath the tree canopy, according to instructions by Massman (1980). Water from the open PVC tubes was conducted to a plastic water bucket and then weighed, to calculate through-fall after each rainfall. In total, six measurements on both slope positions were averaged as a representative value. Stem flow was not measured, on the assumption that it accounts for a negligible proportion of rainfall, as proposed by Schroth et al. (2001), who have investigated various agroforestry systems consisting of trees and crops in a similar fashion to this study.

Overland flow was measured using erosion plots 5.0 m wide and 20.0 m long along the slope. The erosion plots were constructed with cement plates, of which 0.20 m projected above the ground and 0.30 m was inserted into the soil. A tipping bucket system was installed at the lower end of each plot, following the design of Khan and Ong (1997). The water/soil suspension was guided to a bucket and then tipped off into a mesh bag with apertures of <5 μm when the bucket was full. The number of bucket tippings was recorded using an event data logger (Onset Computer Corporation, USA) and then after calibration they were calculated into overland flow volume. A mesh bag was placed in a sediment tank to collect the fine sediments. These were collected after each rainfall from the mesh bag and from the deposition tank, weighed, and then air-dried for chemical analysis. In addition, overland flow water was sampled on an event basis to determine the nitrogen concentrations in different forms.

The soil matric potential was monitored using tensiometers equipped with pressure transducers (26PCDFA6G, Honeywell, USA), which cover a pressure range of ±30.0 psi or ±206.84 kPa and have a very fast response time (<1 ms). A set of tensiometers were installed in a line perpendicular to the direction of slope at 0.20, 0.40, 0.60, 0.85 and 1.50 m depths on both the upper and lower slope positions (Fig. 1). The tensiometers in the agroforestry system were placed 1.50 m away from the tree lines where peanut crops had been sown. Readings of the soil matric potential were recorded at 10-min intervals using a data logger (DL2e, Delta T Inc., UK). Isolines of soil matric potential were drawn using the ordinary kriging method to show seasonal variations in soil moisture within the soil pedons (Zepp, 1991).

High-flow porous ceramic cups were installed at 0.20, 0.40 and 0.85 m depths in the same positions as the tensiometers, to collect soil water samples for the determination of the $NO_3$-N concentration and other chemical properties. Samples were collected once a week, although this failed in the dry season when the soil was too dry. While sampling, a suction pressure of 100 kPa was applied through the sampling bottles one day before.

### 2.3. Chemical analysis

The collected sediment and water samples were immediately sent to the laboratory at the experimental station and stored for analysis. The suspension samples collected from erosion plots were filtered through a 0.45 μm membrane and then used to measure the total nitrogen (TN) concentration following standard procedures (Liu, 1996; Lu, 1999). The filtered water samples were digested with a $K_2S_2O_8$-NaOH solution and then used to determine the TN by ultraviolet spectrophotometry. The soil water samples collected from the cups were used directly to determine the $NO_3$-N concentration by the same method. The air-dried sediment samples collected from the erosion plots were digested with NaOH for TN analysis.

### 2.4. Estimation of subsurface lateral flow and sensitivity analysis

The soil matric potential was used to estimate the components of the water budget by modelling using the Hydrus-2D program (Simunek et al., 1996). As the simulation may fail for numeric reasons if a rainstorm generates saturation excess overland flow along a very long slope, the size of the model domain was kept to 50.0 m in length and 2.0 m in depth, which is equal to the distances between the installed tensiometers at the upper and lower slope positions and the depths of the soil profiles. The slope gradient was 6%. The upper boundary was the atmospheric boundary. The low impermeable parent material with saturated hydraulic conductivity smaller than 0.30 m $d^{-1}$ (Table 1) was approximated by a zero flux bottom boundary. The lower slope boundary was taken as a free drainage boundary and the upper one as a no-flux boundary. The soil layers were distinguished on the basis of the soil bulk density and genetic horizons. These layers were 0–0.25 m, 0.25–1.00 m, and 1.00–2.00 m, and the physical properties were assumed to be uniform within each layer. Saturated hydraulic conductivity ($K_s$) was determined in the laboratory and reported by Jing et al. (2008). The parameters of the van Genuchten function $\theta(\psi_m)$ were derived from the best fit using the Rosetta pedotransfer function (Schaap et al., 2001), and were embedded in Hydrus-2D and used as the initial values for an inverse calibration. This inverse calibration was performed against soil matric potential measured in the field, with fixed saturated hydraulic conductivity ($K_s$) measured by Jing (2004) (Table 1).

Assuming no flux through the lower base boundary, water moving below the root zone was then redirected through this as a subsurface lateral flow. Therefore, annual subsurface lateral flow ($Q_{inter}$) was taken in the water budget Eqn. (1) as follows:

$$Q_{inter} = P - (R + ET + E_{Int} + \Delta S) \quad (1)$$

where $P$ is the measured precipitation; $R$ is the measured overland flow; ET is the simulated actual evapotranspiration; $E_{Int}$ is the estimated or measured precipitation interception; $\Delta S$ is the calculated change in soil water storage.

The actual evapotranspiration, ET, was simulated based on the potential evapotranspiration, following the FAO Penman-Monteith method (Allen et al., 1998), and the water uptake reduction module following the Feddes model (Feddes et al., 1978) embedded in Hydrus-2D. In the Feddes model plant phenology, root distribution, canopy size and water stress in relation to soil moisture regimes were considered. Precipitation interception in the agroforestry system was estimated by subtracting measured through-fall from total precipitation. It was estimated for the mono cropping system according to the method of Hoyningen-Huene (1983). This method was developed for determining rainfall interception in the canopy of crops and has been applied to many hydrological models (Nunes et al., 2006; Kozak et al., 2007).

When $P < P_{gr}$,

$$E_{int} = -0.42 + 0.245 \times P + 0.2 \times \mathrm{LAI} - 0.011 \times P^2 + 0.0271 \times P \times \mathrm{LAI} - 0.0109 \times \mathrm{LAI}^2 \quad (2)$$

When $P > P_{gr}$,

$$E_{int} = 0.935 + 0.498 \times \mathrm{LAI} - 0.00575 \times \mathrm{LAI}^2 \quad (3)$$

$$P_{gr} = 11.05 + 1.223 \times \mathrm{LAI} \quad (4)$$

where $P$ is the precipitation measured in an open area; $P_{gr}$ is the maximum precipitation interception at maximum LAI.

**Table 1**
Selected chemical and phycicial properties and the best fitted parameters of van Genuchten water retention curve by soil horizon at both the upper and lower slope positions in the agroforestry system and the mono cropping system.

| Land use | Slope position | Soil horizon[a] | Depth (m) | Bulk density ($mg\,m^{-3}$) | <0.002 mm clay ($g\,kg^{-1}$) | pH | Soil organic C ($g\,kg^{-1}$) | Total N ($g\,kg^{-1}$) | $K_s$[b] ($m\,d^{-1}$) | van Genuchten parameters of $\theta(\psi_m)$[c] | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | $\theta_r$ ($m^3\,m^{-3}$) | $\theta_s$ ($m^3\,m^{-3}$) | $A$ ($m^{-1}$) | $n$ |
| Agroforestry system | Upper slope | Ap | 0–0.25 | 1.26 | 339 | 4.66 | 16.41 | 0.95 | 2.42 | 0.19 | 0.5 | 2.5 | 1.455 |
| | | AB | 0.25–0.50 | – | 400 | 4.33 | 2.96 | 0.34 | 0.60 | 0.221 | 0.44 | 2.1 | 1.424 |
| | | Bt | 0.50–1.00 | 1.45 | – | 4.24 | 2.44 | 0.32 | | | | | |
| | | BCv | 1.00–1.30 | 1.52 | 409 | – | – | – | 0.27 | 0.189 | 0.434 | 2.1 | 1.312 |
| | | | 1.30–2.00 | – | – | – | – | – | | | | | |
| | Lower slope | Ap | 0–0.25 | 1.39 | 324 | 5.06 | 10.85 | 0.68 | 2.46 | 0.191 | 0.5 | 2.2 | 1.455 |
| | | B1 | 0.25–0.50 | 1.44 | 307 | 4.63 | 2.705 | 0.325 | 0.50 | 0.22 | 0.44 | 2.0 | 1.414 |
| | | B2t | 0.50–0.90 | 1.49 | 380 | 4.18 | 2.59 | 0.28 | | | | | |
| | | B3 | 0.90–1.30 | 1.57 | 414 | – | – | – | 0.25 | 0.189 | 0.436 | 1.2 | 1.315 |
| | | | 1.30–2.00 | – | – | – | – | – | | | | | |
| Mono cropping system | Upper slope | Ap | 0–0.25 | 1.23 | 315 | 5.06 | 14.43 | 0.8 | 2.40 | 0.196 | 0.5 | 2.2 | 1.455 |
| | | AB | 0.25–0.50 | – | 393 | 4.89 | 3.33 | 0.31 | 0.64 | 0.229 | 0.425 | 2.5 | 1.434 |
| | | Bt | 0.50–1.00 | 1.43 | 404 | 4.42 | 2.75 | 0.32 | | | | | |
| | | BCv | 1.00–1.30 | 1.51 | 450 | – | – | – | .030 | 0.249 | 0.434 | 1.8 | 1.325 |
| | | | 1.30–2.00 | – | – | – | – | – | | | | | |
| | Lower slope | Ap | 0–0.25 | 1.35 | 313 | 5.19 | 11.22 | 0.62 | 1.98 | 0.19 | 0.5 | 2.5 | 1.455 |
| | | Bt1 | 0.25–0.50 | 1.46 | 358 | 4.94 | 2.9 | 0.27 | 0.70 | 0.221 | 0.44 | 2.1 | 1.424 |
| | | Bt2 | 0.50–0.90 | 1.51 | 353 | 4.24 | 2.44 | 0.25 | | | | | |
| | | Bt3 | 0.90–1.30 | 1.57 | 341 | – | – | – | 0.26 | 0.172 | 0.434 | 1.2 | 1.32 |
| | | | 1.30–2.00 | – | – | – | – | – | | | | | |

[a] The small letters for soil horizons are: t, accumulation of silicate clay; and v, plinthite.
[b] $K_s$, saturated hydraulic conductivity, determined in laboratory by Jing (2004).
[c] $\theta(\psi_m)$, water retention curve.

Subsurface lateral flow was calculated from the simulated water budget, which was assessed by comparing the simulated and measured soil matric potential and by analyzing the sensitivity of subsurface lateral flow to the (not measured) input parameters. The goodness of fit between the simulated and the measured soil matric potential was assessed using a relative error of geometric mean (RGME, Eq. (5)) and a standard deviation (SPE, Eq. (6)) of mean percentile error (MPE, Eq. (7)). It was optimized with an RGME approaching one and an SPE of zero (Morgenstern and Kloss, 1995; Zepp and Belz, 1992). In order to avoid a zero denominator, the denominator was converted into soil water potential by setting the soil surface at zero gravity potential (negative, downward).

$$\mathrm{RGME} = \sqrt{\frac{S_1}{M_1} \times \frac{S_2}{M_2} \cdots \frac{S_n}{M_n}} \tag{5}$$

$$\mathrm{SPE} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(\mathrm{MPE} - \frac{|M_i - S_i|}{M_i} \times 100\right)^2} \tag{6}$$

with

$$\mathrm{MPE} = \frac{1}{n}\sum_{i=1}^{n}\frac{|M_i - S_i|}{M_i} \times 100 \tag{7}$$

where $S_i$ and $M_i$ are the simulated and measured soil water potentials respectively and $n$ is the total of these pairs.

The simulation was sensitive to bulk stomatal resistance and could be affected by the estimated precipitation interception by peanut crop and the best fitting hydraulic parameters of the van Genuchten function listed in Table 1. Therefore, the sensitivity of subsurface lateral flow to model parameters was analyzed by modifying by ±20% of the best fitting values of bulk stomatal resistance as well as the precipitation interception by peanut crop, and by exchanging the hydraulic parameters between the two cropping systems.

### 2.5. Estimation of nitrogen losses

The annual $NO_3$-N loss through subsurface lateral flow ($N_{\mathrm{inter}}$, kg ha$^{-1}$) was estimated by multiplying the $NO_3$-N concentration in soil water by the subsurface lateral flow flux during the sampling intervals (Eq. (8)). Similar methods have been used in other studies (Jansons et al., 2003; Alva et al., 2006).

$$N_{\mathrm{inter}} = \sum_{i=1}^{k} C_i \times q_{\mathrm{inter}} \times \Delta t_i \times 10 \tag{8}$$

where $k$ represents the sampling times in one monitoring year; $C_i$ is the $NO_3$-N concentration (kg m$^{-3}$) at the $i$th sampling time; $q_{\mathrm{inter}}$ is the subsurface lateral flow flux (mm d$^{-1}$) at the $i$th sampling time; $\Delta t_i$ is the time interval (d) between the $i$th and $(i+1)$th sampling time.

Similarly, annual total nitrogen loss through overland flow and sediments for each rainfall event was calculated by multiplying the amount of overland flow and sediments by the TN concentration in them.

## 3. Results

### 3.1. Soil properties

Selected physical, hydraulic and chemical properties within soil pedons are presented in Table 1. They varied between the two slope positions, but not between the two cropping systems. Although the clay content was no more than 20% more in the Bt horizon than in the overlying AB horizon, the Bt horizon has been defined here because it displayed typical features of clay coatings on semi-angular aggregates and colour changes. The clay content increased with depth on the upper slope position, but this was not observed on the lower slope. The bulk density and saturated hydraulic conductivity also increased with depth and were not differentiated between the two cropping systems. From all the horizons, the saturated hydraulic conductivity was highest in the Ap horizons (2.00–2.50 m d$^{-1}$) and lowest in the Bt/BCv horizons (<0.30 m d$^{-1}$). The organic carbon and TN content were highest in the Ap horizon, and lower on the lower slope positions than on the upper positions in both cropping systems.

### 3.2. Soil water regime along the slope

The soil matric potential is exemplified in the rainstorm event of the 7th of April 2003, when the precipitation was 48.0 mm (Fig. 2). The soil pedons before the storm were wet, with a soil matric potential ranging from −8.0 to −1.0 kPa. During the rainstorm the soil matric potential fluctuated with the rainfall intensity and was negative all the time at the upper soil depths, from 0.20 to 0.60 m in both cropping systems, which means there was no water saturation. After the rainstorm, the soil matric potential became more negative over time at the upper soil depths on both slope positions, and showed slight decreases at the lower depths on the upper slope. On the lower slope, the soil matric potential at a depth of 1.50 m became positive in both cropping systems, indicating soil water saturation. The maximum perched water table at the 1.5 m depth was greater (0.32 m vs. 0.17 m) and it took longer to reach the maximum (1020 min vs. 730 min) in the agroforestry system than in the mono cropping system.

The isolines of soil matric potential showed distinct temporal and spatial variations of soil moisture along the pedons and between the cropping systems (Fig. 3). During the rainy season, the pedons were saturated in a larger domain on the lower slope than on the upper slope, and the difference in the saturated domains between the upper and lower slope positions was more profound in the mono cropping system than in the agroforestry system. Soon after the rainy season, the soil pedons became drier, going from the surface to the deeper soil layers, and the dry zone extended to a larger area in the deep soil layers in the agroforestry system than in the mono cropping system. The minimum and median of soil matric potential, particularly in the dry season (Fig. 4), illustrate that soil moisture varied more strongly in the shallower soil depth in the mono cropping system than in the agroforestry system (0.40 m vs. 1.50 m), indicating a deeper active root zone taking up soil water in the agroforestry system than in the mono cropping system.

### 3.3. Spatial and temporal dynamics of $NO_3$-N concentration in soil water

Since soil water sampling using porous suction cups did not function in dry conditions, the $NO_3$-N concentration presented here, in Fig. 5, is only for those sampled in the relatively wet soil conditions over 94 weeks during the three years of monitoring. The $NO_3$-N concentration had greater temporal and spatial variations at all soil depths in the mono cropping system than in the agroforestry system. In the latter, the $NO_3$-N concentration was generally lower than 10 mg L$^{-1}$ at all depths and had no seasonal variation except for a few scattered higher values on the upper slope in the year 2005. Although the $NO_3$-N concentration decreased at the 0.85 m depth from 2003 to 2005, the average was greater at the 0.85 m depth than at the 0.20 and 0.40 m depths on both slope positions (4.3–7.6 mg L$^{-1}$ vs. 0.6–2.3 mg L$^{-1}$) and was lower at all soil depths on the lower slope than on the upper slope (1.9 mg L$^{-1}$ vs. 4.2 mg L$^{-1}$).

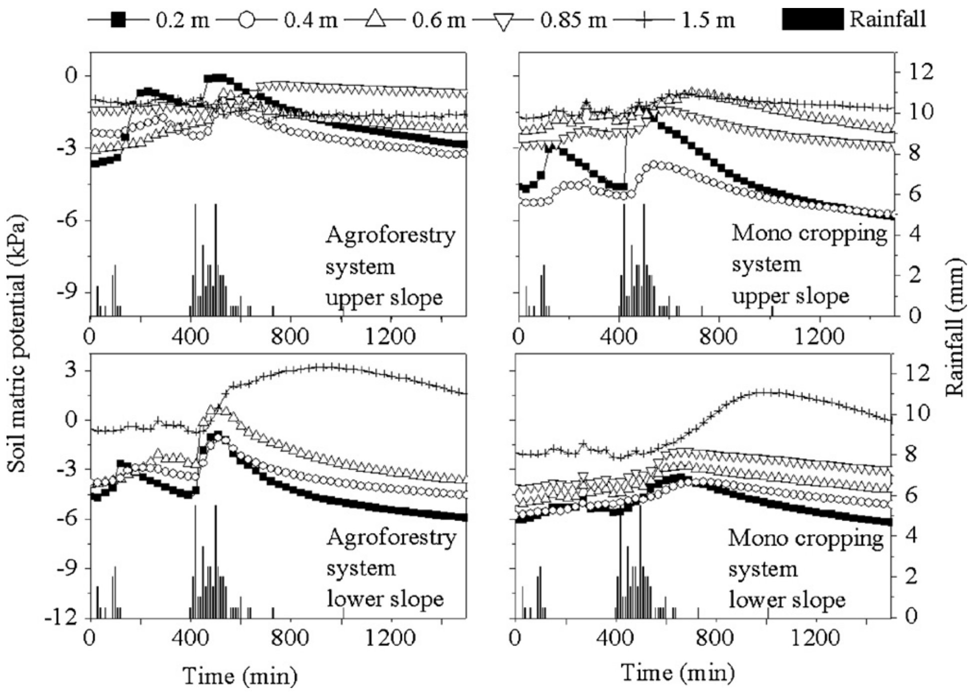


**Fig. 2.** Dynamics of soil matric potential at different depths in the agroforestry system (left column) and the mono peanut cropping system (right column) during and after the rainstorm of 7th April, 2003. Missing depths are due to a malfunction of the tensiometers.

In the mono cropping system the $NO_3$-N concentration presented two peaks at all the depths in each monitoring year except for the 0.85 m depth on the upper slope (Fig. 5). The first peak occurred during the period from middle May to late June and the second during the period from October to the following February. The peaks ranged from 18.6 to 52.5 mg $L^{-1}$ at the 0.20 m depth, from 14.1 to 32.8 mg $L^{-1}$ at the 0.40 m depth and from 16.1 to 26.0 mg $L^{-1}$ at the 0.85 m depth. The first peak was larger than the second at the comparable depth. The peak size decreased with increasing soil depth at both slope positions in the relative seasons and was greater on the lower slope than on the upper slope at the comparable soil depths. In addition, the peak size at the 0.20 and 0.40 m depths increased with time, with a greater increase on the lower slope than on the upper slope. There were time lags between the peaks in the same season between the shallower and deeper soil depths. For example, the peak $NO_3$-N concentration at the 0.20 m depth appeared on the 31st May 2004, 24 days after the basal fertilization, while the peak concentration at 0.85 m appeared on the 15th July 2004, 90 days after basal fertilization. This time lag corresponds to an average transport velocity of 5.2 m $a^{-1}$. The minimum $NO_3$-N concentration between the two peaks in each monitoring year was as low as 0–1.6 mg $L^{-1}$ at all soil depths, with no differences in its average among the soil depths at both slope positions.

### 3.4. Estimation of subsurface lateral flow and associated nitrate loss

The simulated and measured soil water potentials fitted well together. This was exemplified in a visual comparison for the 0.40, 0.85 and 1.50 m depths on the upper slope position (Fig. 6), and verified by the RGME and SPE data calculated for the three-year period (Table 2). The RGME ranged from 1.01 to 1.46 and the SPE from 24 to 141. The smaller values indicate that the simulation was better at the 0.85 and 1.50 m depths than at the 0.40 m depth, although the simulation did not match well with the sharp decrease in soil water potential during the transit period from the rainy season to the dry season at the 0.85 and 1.50 m depths.

A sensitivity analysis showed that the simulation was most sensitive to the bulk stomatal resistance in the Penman-Monteith function and less sensitive to the estimated precipitation interception by peanut crops and the best fitting soil hydraulic parameters (Table 3). The variation in bulk stomatal resistance by ±20% resulted in a change in the subsurface lateral flow from −6.4% to 11.2% in the agroforestry system and from −7.7% to 4.3% in the mono peanut crop. The variation by ±20% in the estimated precipitation interception caused a variation from −1.5% to 1.5% in both cropping systems. Exchanging the hydraulic parameters with the best fit between the two cropping systems resulted in a variation of only −3.7% to 0.7%.

The simulated water budget (Table 4) showed that in each year the mono cropping system had greater subsurface lateral flow and evapotranspiration and less precipitation interception than the agroforestry system. In the mono cropping system, the subsurface lateral flow accounted for 35% to 42% of the annual rainfall and the evapotranspiration for 35.5% to 48.9%. In the agroforestry system the subsurface lateral flow accounted for 14% to 34% of annual rainfall and the evapotranspiration for 45.2% to 65.7%. It was noted that the difference in interception between the two cropping systems was approximately equivalent to the difference in subsurface lateral flow.

The estimate of $NO_3$-N loss through subsurface lateral flow was larger in the mono cropping system than in the agroforestry system (Table 5). The $NO_3$-N loss accounted for 30.6% to 40.0% of the total N fertilizer applied in the mono cropping system and for 9.8% to 31.0% in the agroforestry system. The $NO_3$-N loss was 2 to 9.5 times as much as the TN loss through soil erosion with overland flow. The differences in $NO_3$-N loss through subsurface lateral flow and total N loss between the two cropping systems were small in 2003, but grew in the following two years, being much larger in the mono cropping system than in the agroforestry system.

## 4. Discussion

Intensive agriculture imposes environmental risks as a result of the excessive nutrients left in soils and ground water (Groffman,

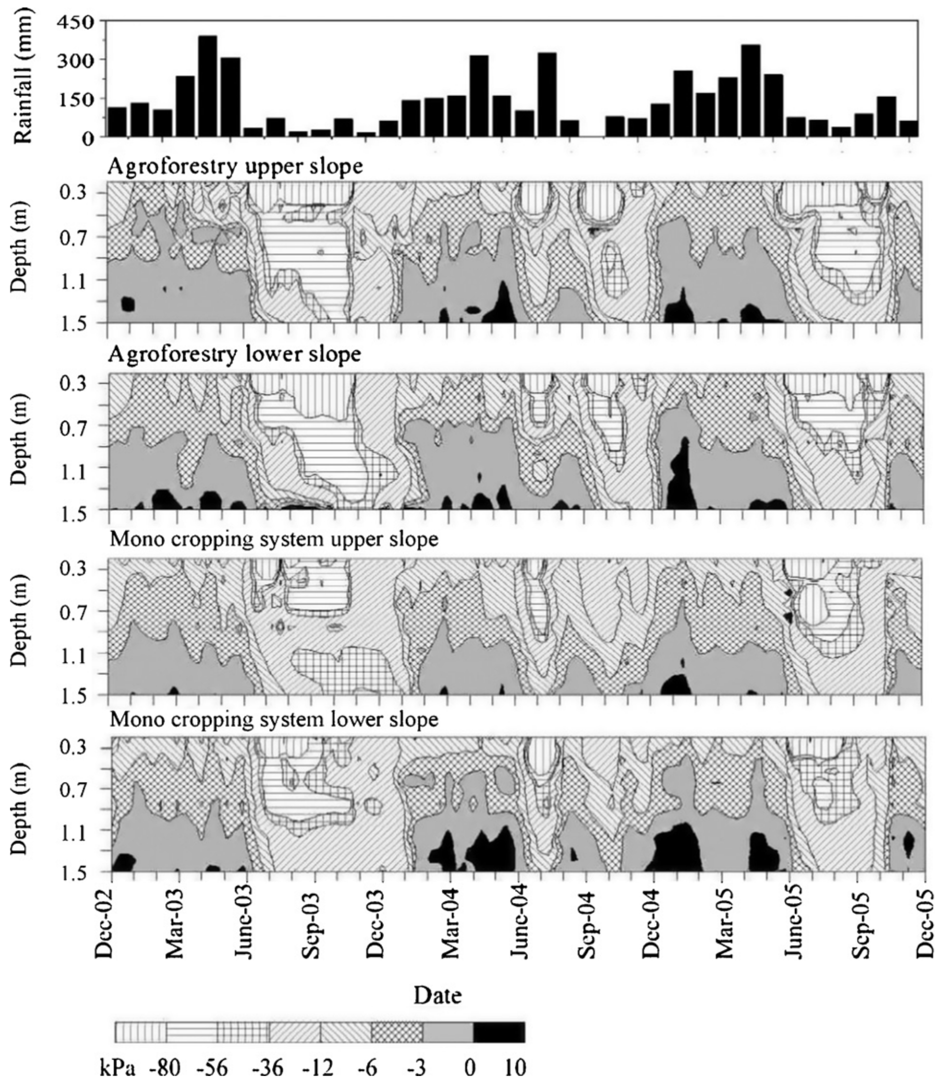


**Fig. 3.** Soil moisture regime interpolated by ordinary kriging from the soil matric potential measured at the 0.20, 0.40, 0.60, 0.85 and 1.50 m depths on the upper and lower slope positions in the agroforestry system and the mono peanut cropping system, from 2003 to 2005.

2000; Jalali, 2005). Although we did not replicate this in the field, so as to reduce the costs of instruments and sampling, the three-year monitoring period allowed a comparison between the dynamics of the two cropping systems, and its reproducibility in the dynamics of nitrate profiles revealed significant differences between the two cropping systems. The nitrogen applied exceeded crop uptake in the mono peanut system and leached down to a depth of 0.85 m under the subtropical monsoonal weather, causing maximum $NO_3$-N concentration in soil water within the soil profile: it ranged from 14 to 52 mg $L^{-1}$ and was much greater than the

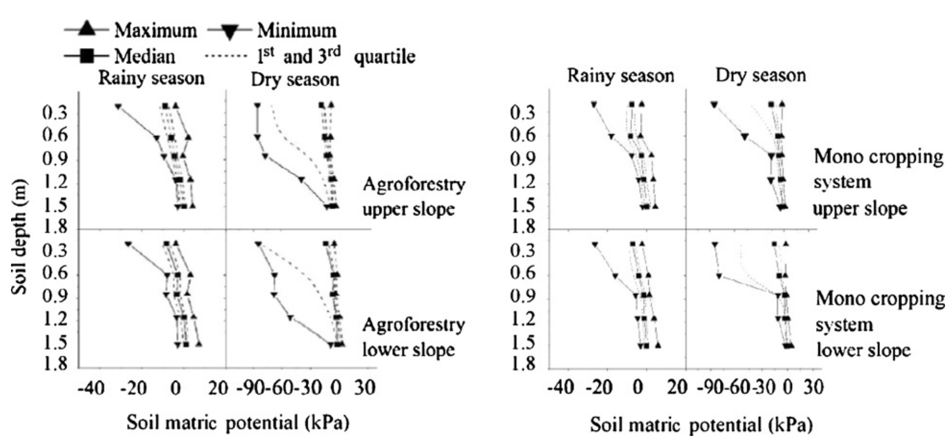


**Fig. 4.** Vertical distribution of median, maximum and minimum of soil matric potential for the rainy season from March to July and the dry season from July to November on the upper and lower slope positions in the agroforestry system (left) and the mono peanut cropping system (right).

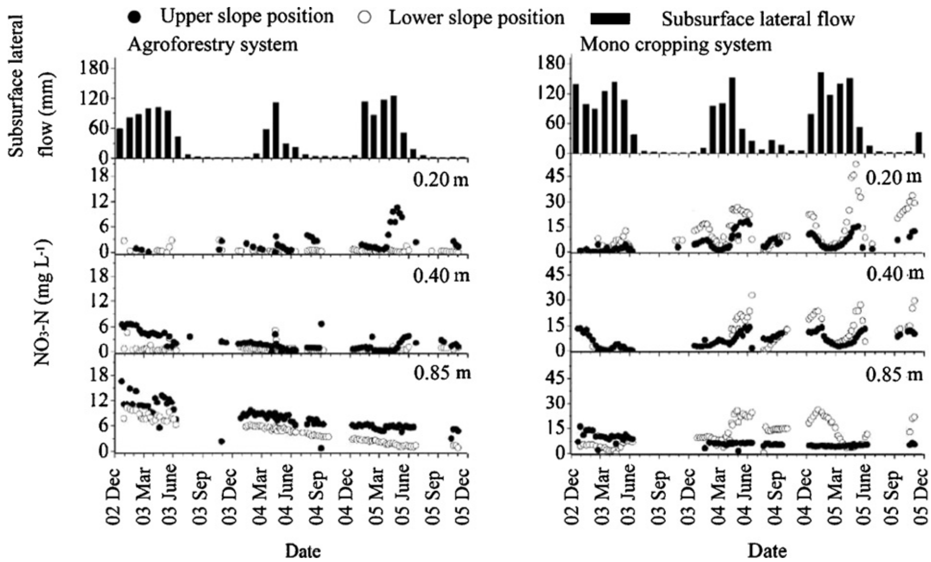


**Fig. 5.** Dynamics of $NO_3$-N concentration and subsurface lateral flow along the slope in the agroforestry system (left) and the mono peanut cropping system (right) from 2003 to 2005.

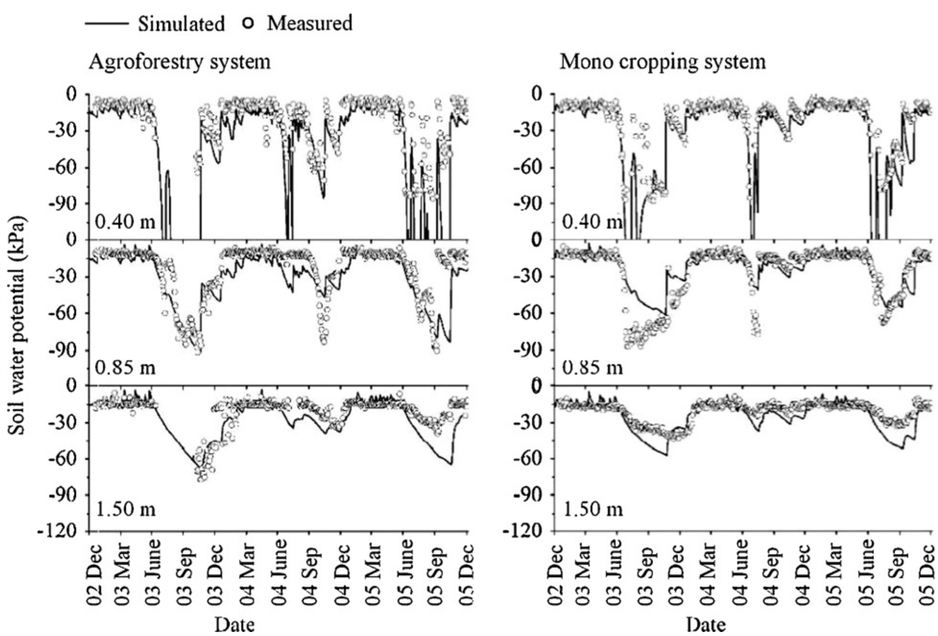


**Fig. 6.** Measured and simulated soil water potential (soil matric potential plus gravity potential) on the upper slope position in the agroforestry system (left) and the mono peanut cropping system (right).

limit for drinking water (10 mg L$^{-1}$) set by the World Health Organization. Although more nitrogen was applied in the agroforestry system, the $NO_3$-N concentration in soil water was much lower (0.6–7.6 mg L$^{-1}$) there than in the mono cropping system. Therefore, an agroforestry system consisting of citrus and crop would reduce the environmental risk of $NO_3$-N pollution through subsurface lateral flow.

The $NO_3$-N concentration varied strongly with the season in the mono cropping system (Fig. 6). This is consistent with other studies of different cash crops and subsistence cereals (see e.g. Garg et al.,

**Table 2**
Relative geometrical mean error (RGME) and standard deviation (SPE) of soil water potential at different soil depths between the measured and simulated using Hydrus-2D during the period from 2003 to 2005 on the upper and lower slope positions in the agroforestry system and the mono cropping system.

| Cropping system | Slope position | RGME | | | SPE | | |
|---|---|---|---|---|---|---|---|
| | | 0.40 m | 0.85 m | 1.50 m | 0.40 m | 0.85 m | 1.50 m |
| Agroforestry system | Upper slope | 1.39 | 1.4 | 1.17 | 141 | 80 | 47 |
| | Lower slope | 1.16 | 1.04 | 0.98 | 65 | 48 | 31 |
| Mono cropping system | Upper slope | 1.46 | 1.01 | 0.92 | 114 | 22 | 34 |
| | Lower slope | 1.4 | 1.08 | 1.26 | 87 | 29 | 53 |

**Table 3**
Simulated subsurface lateral flow ($Q'_{inter}$) in the agroforestry system and the mono cropping system under the simulation parameter change scenarios (±20% change in the best fitted parameters of bulk stomatal resistance and precipitation interception by peanut crop and exchange of soil hydraulic parameters (see Table 1 between the two cropping systems) and its relative variation to the best fitted $Q_{inter}$ reported in Table 4 during the period from 2003 to 2005.

| Change scenario | Year | Agroforestry system | | Mono cropping system | |
|---|---|---|---|---|---|
| | | $Q'_{inter}$(mm) | Relative variation[a] (%) | $Q'_{inter}$(mm) | Relative variation[a] (%) |
| Bulk stomatal resistance | | | | | |
| +20% | 2003 | 535 | 3.8 | 654 | 2.7 |
| | 2004 | 259 | 11.2 | 431 | 4.3 |
| | 2005 | 578 | 5.1 | 716 | 0.4 |
| −20% | 2003 | 495 | −3.9 | 621 | −2.5 |
| | 2004 | 218 | −6.3 | 381 | −7.7 |
| | 2005 | 515 | −6.4 | 671 | −5.9 |
| Precipitation interception by peanut crop | | | | | |
| +20% | 2003 | 511 | −0.7 | 628 | −1.4 |
| | 2004 | 229 | −1.7 | 410 | −0.7 |
| | 2005 | 545 | −0.9 | 705 | −1.1 |
| −20% | 2003 | 519 | 0.8 | 647 | 1.5 |
| | 2004 | 236 | 1.5 | 415 | 0.4 |
| | 2005 | 555 | 0.9 | 720 | 1.0 |
| Hydraulic parameters | | | | | |
| Exchanged between the two cropping systems | 2003 | – | – | 642 | 0.7 |
| | 2004 | – | – | 409 | −1.0 |
| | 2005 | – | – | 698 | −2.1 |
| | 2003 | 511 | −0.7 | – | – |
| | 2004 | 224 | −3.7 | – | – |
| | 2005 | 540 | −1.8 | – | – |

[a] The relative variation is defined as the difference between the simulated value under a change scenario and the best fitting (Table) divided by the simulated value under the best fitting.

**Table 4**
Annual water budget items estimated using Hydrus-2D for the agroforestry system and the mono cropping system during the period from 2003 to 2005.

| Year | Cropping system | Water budget items[a] (mm) | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | $P$ | $E_{int}$ | $R$ | ET | $Q_{inter}$ | $\Delta S$ | $Q_{inter}/P$% |
| 2003 | Agroforestry system | 1531 | 214 | 182 | 804 | 515 | −184 | 33.6 |
| | Mono cropping system | | 55 | 223 | 748 | 637 | −133 | 41.6 |
| 2004 | Agroforestry system | 1632 | 208 | 106 | 1007 | 233 | 79 | 14.3 |
| | Mono cropping system | | 65 | 266 | 785 | 413 | 104 | 25.3 |
| 2005 | Agroforestry system | 1882 | 223 | 226 | 852 | 550 | 32 | 29.2 |
| | Mono cropping system | | 73 | 429 | 669 | 713 | −3 | 37.9 |

[a] $P$, precipitation measured by rain gauge in the field; $R$, overland flow measured by erosion plots; ET, evapotranspiration calculated using Hydrus-2D; $E_{int}$, rainfall interception measured in the field for trees canopy or that calculated using Hoyningen-Huene method for peanut crops; $Q_{inter}$, subsurface lateral flow calculated using Hydrus-2D; $\Delta S$, change in soil water storage calculated using Hydrus-2D.

2005; Honisch et al., 2002). The seasonal variations in the mono cropping system can be attributed to the seasonal transformation of N fertilization and the vertical leaching of $NO_3$-N through the soil profile. In this study two peaks were observed, one appearing at the 0.20–0.85 m depth, particularly on the lower slope position during the period from middle May to late June, and the other at the depths of 0.20 and 0.40 m during the period from October to February, when no crops were grown in the field. The variation in the soil water regime suggested that the active root zone of the peanut crop was limited to a depth of 0.40 m (Fig. 4). Therefore, the first peak of $NO_3$-N concentration at the 0.85 m depth on the lower slope position indicated that the application of fertilizer at a rate of 161.0 kg N $ha^{-1}$ $a^{-1}$ exceeded the crop's nitrogen demand, and so the $NO_3$-N released from urea in the upper soil layers leached and accumulated in the deeper soil layer. Although no peaks appeared at the 0.85 m depth on the upper slope, the high concentration at the 0.85 m depth (11.7 mg $L^{-1}$) illustrated that $NO_3$-N leached from the upper soil layer could not be used by the peanut crop and would accumulate in the deeper soil. The degree of $NO_3$-N concentration was greater on the lower slope than on the upper slope. Assuming

**Table 5**
Annual nitrogen loss through different hydrological pathways during the period from 2003 to 2005 in the agroforestry system and the mono cropping system.

| Year | Land use | Soil loss (mg $ha^{-1}$) | Annual nitrogen loss (kg $ha^{-1}$ $a^{-1}$) | | | |
|---|---|---|---|---|---|---|
| | | | Sediment | Overland flow | Subsurface lateral flow | Total |
| 2003 | Agroforestry system | 1.49 | 1.18 | 4.16 | 48.65 | 53.99 |
| | Mono cropping system | 2.63 | 2.22 | 5.96 | 45.89 | 54.07 |
| 2004 | Agroforestry system | 0.29 | 0.2 | 2.19 | 15.48 | 17.87 |
| | Mono cropping system | 0.49 | 0.53 | 6.88 | 40.34 | 47.75 |
| 2005 | Agroforestry system | 1.83 | 3.05 | 6.25 | 19.74 | 29.04 |
| | Mono cropping system | 3.62 | 4.08 | 13.71 | 64.31 | 82.1 |

that there was no difference in N demand by the same crop on the different slope positions, this suggested that the $NO_3$-N accumulated in the deeper layers did not result only from vertical leaching or preferential flow, but also from subsurface lateral transport. The time lag between the peaks of $NO_3$-N concentration at adjacent depths from 0.20 to 0.85 m demonstrated that the average nitrate transport velocity was as fast as 5.2 m $a^{-1}$ in 2004, which was much faster than that (1.2 m $a^{-1}$) in a coarse loamy agricultural soil reported by Honisch et al. (2002). Strong preferential flow occurred during storms, as indicated by the soil water potential, which was always negative in the surface soil during a storm, but was positive in the deep soil layers at the same time (Fig. 2). In the mono cropping system, the $NO_3$-N concentration decreased very quickly after the first peak in each year. This could be explained by more active peanut crop uptake, stronger lateral transport during storms or by enhanced denitrification owing to wetter soil conditions after the first peak (from early June). Denitrification is inhibited when soil pH and organic carbon are low. Richards and Colin (1999) reported that the subsoil of agricultural fields with low soil organic carbon content (ranging from 1.8 to 2.6 g $kg^{-1}$) largely inhibited the release of $N_2O$ (from 4.2 to 28.4 μg $N_2O$-N $kg^{-1}$ $d^{-1}$). The soils studied had low pH and organic carbon content below the 0.85 m depth (Table 1). Therefore, the simultaneous decrease in $NO_3$-N concentration at the 0.85 m depth on the lower slope can be explained only by the generation of lateral transport, because the active root zone was limited there. This observation supports the assumption made by Tang et al. (2008) that spatial and temporal variations in the chemistry of water sources within the same studied catchment were attributable to subsurface lateral flow. The second peaks in the mono cropping system were observed only within the 0.40 m depth and they were smaller than the first peaks. This indicated a slower N transformation and transport during the dry season from October to February. The following decreases lasted from March when the rainy season started to early May when fertilizers were applied. Since there was no crop coverage during that period, the decrease in $NO_3$-N concentration in the deeper soil layers, down to the 0.40 m depth, could also be attributed to subsurface lateral flow.

The $NO_3$-N concentration in the agroforestry system was much lower and had less seasonal variability than that in the mono cropping system (Fig. 5). Given the similar amount of N fertilizers applied in both cropping systems, the lower $NO_3$-N concentration in soil water was probably to be attributed to the differences in N inputs through biological fixation by peanut crops and in uptake by different plants. Biological N fixation by the peanut crop falls in a range from 22 to 200 kg N $ha^{-1}$ in shoot, with a mean of 100 kg N $ha^{-1}$ in shoot in South-East Aisa (Peoples et al., 2009). Kuenyler and Craig (1986) reported a similar figure of fixed N by peanut crop (112 kg $ha^{-1}$ $a^{-1}$) at a low N application rate and this figure was used to estimate N budget at watershed scale in southern China (Yan et al., 1999). Given that N fixation by peanut crop was 112 kg $ha^{-1}$ $a^{-1}$, the agroforestry system would have lower than 70 kg-N $ha^{-1}$ $a^{-1}$ N input by biological fixation than the mono cropping system considering the difference in the effective area occupied by peanut crops. The actual amount N fixation by peanut crop would be much lower in our case when as much as 161 kg-N $ha^{-1}$ $a^{-1}$ of mineral N was applied to peanut crop. Application of large amounts of chemical N will inhibit biological N fixation because of the reduction in the number of nodules and nitrogenase activity (Dakora, 1998; Zhao et al., 2007). Zhao et al. (2007) reported that compared with the no mineral N treatment and the recommended fertilization treatment (90 kg-N $ha^{-1}$ $a^{-1}$), the nitrogenase activity in the treatment with local fertilization (180 kg-N $ha^{-1}$ $a^{-1}$) decreased by 64% and 38%, respectively. Therefore, the total N inputs would be lower by about 0 to 70 kg-N $ha^{-1}$ $a^{-1}$ in the agroforestry system than in the mono cropping system. This could be one of the important reasons that the $NO_3$-N concentration in soil water in the agroforestry system. Another important reason was that the deeper root system may have taken up the $NO_3$-N leached to the deep soil layers. The stronger variations in the water regime within soil profile (Fig. 4) suggested that the root zone extended to a depth of 1.50 m in the agroforestry system but only to 0.40 m in the mono cropping system. This was confirmed by a field excavation which showed that 90% of the roots of citrus trees were concentrated within a depth of 0.85 m, and 88% of peanut crop roots within 0.20 m (data not shown). The tree roots in agroforestry systems can extend to adjacent areas occupied by crops, leading to an increase in root density there (Rao et al., 1993; Nair et al., 2007). The scattered high $NO_3$-N concentration at the 0.20 and 0.40 m depths in the agroforestry system (Fig. 5) indicated that $NO_3$-N released in the surface soil under a peanut cropping alley passed by the root zone, probably through preferential flow. The leached $NO_3$-N was then taken up by the extended root zone of the trees, resulting in a low $NO_3$-N concentration and no peaks at any depths down to 0.85 m in the agroforestry system. The $NO_3$-N concentration at the 0.85 m depth decreased between 2003 and 2005; this was probably due to the downward growth of citrus tree roots over time, which took up the accumulated $NO_3$-N in the deep soil layers.

The spatial variation in the soil water regime (Figs. 2 and 3) demonstrates the subsurface lateral flow through the soil pedon along the slope. The positive soil matric potential at the 1.50 m depth (Fig. 2) indicated that the soil water was saturated above 1.50 m on the lower slope position after the storm in both cropping systems. A much higher saturated hydraulic conductivity in the upper soil layers led to the quick formation of a perched water table above 1.50 m, where the hydraulic conductivity was low (Table 1). The water table further rose to a height of 0.32 m in the agroforestry system and 0.17 m in the mono cropping system, and it took longer to reach the maximum height in the agroforestry system than in the mono cropping system (1020 min vs. 730 min). The rise in the soil water table after the storm did not result from a rise in the groundwater table, as the tensiometer measurements were conducted far above the stream and pond water level. Our previous study had demonstrated a periodic drying up of streams during the antecedent rainless period and dry seasons of autumn and winter (Tang et al., 2008), confirming a negligible influence of the groundwater table. Therefore, the decrease in the water table depth at 1.5 m demonstrated that 'old water' might be propelled out of the soil layer (Collins et al., 2000), generating a considerable volume of subsurface lateral flow along the slope. The height of the water indicated that the agroforestry system had a greater water retention capacity in the soil pedon than the mono cropping system. The increase in water retention capacity is referred to in the literature as the reservoir function of trees (Myers, 1983; Bruijnzeel, 2004). This reservoir function can be ascribed to a tree root system that modifies a soil's hydraulic characteristics and to the tree canopy that intercepts rainfall and reduces actual rainfall intensity (Asdak et al., 1998; Bruijnzeel, 2004). The reservoir function in the agroforestry system was confirmed by a long-term monitoring of the soil water regime, which demonstrated that soil profiles in the rainy season were wetter in the agroforestry system than in the mono cropping system (Fig. 3). The increased soil water retention capacity suggested that the subsurface lateral flow generated from the agroforestry system would be smaller than from the mono cropping system.

Subsurface lateral flow was calculated from the simulated water budget using Hydrus-2D. Although the RGME was greater than one and the SPE far greater than zero (Table 2), this indicated that the simulation was not perfect. However, it was acceptable, as a simulation can be more sensitive to soil matric potential than to soil water content. Morgenstern and Kloss (1995) suggested that an RGME ranging from 0.87 to 1.88 for simulated soil matric potential corresponded to an RGME from 0.94 to 1.08 for simulated soil water content, while an SPE ranging from 14 to 674 for simulated soil matric potential corresponded to an SPE from 4 to 34 for simulated soil water content. In addition, as Fig. 6 depicts, the mismatch occurred in the surface soils and during the dry season. It would be expected that this would have little influence on the quantification of subsurface lateral flow because the latter is normally generated from the deep soil layers during the rainy season. The sensitivity analysis (Table 3) indicated that the calculated subsurface lateral flow was sensitive to the stomatal resistance embedded in the Penman-Monteith function and to the precipitation interception by peanut crop estimated using the Hoyningen-Huene method. However, this did not conceal the differences between the two cropping systems. A small decrease in bulk stomatal resistance results in an increase in actual evapotranspiration, and thereby in a decrease in subsurface lateral flow. Even if the bulk stomatal resistance decreased by 20% in the agroforestry system and increased by 20% in the mono cropping system, the subsurface lateral flow would still be much greater in the mono cropping system than in the agroforestry system. The estimated change of ±20% in precipitation intercept by the peanut crop had little influence on the subsurface lateral flow in either land use system and the subsurface lateral flow was always greater in the mono cropping system than in the agroforestry system. In addition, even if there had been a systemic error in calculating the potential evapotranspiration and precipitation interception by the peanut crop, the calibration of the model could have covered the errors by the best fit of the van Genuchten parameters. By exchanging the best fitting hydraulic parameters between the two cropping systems, there was also a negligible influence on the subsurface lateral flow (Table 3). All the sensitivity analyses indicated that the simulation was reliable, showing a greater subsurface lateral flow in the agroforestry system than in the mono cropping system (Table 4).

The water budget demonstrated that the sum of interception, evapotranspiration and subsurface lateral flow accounted for 91.7% of annual rainfall in the agroforestry system and 82.9% in the mono cropping system respectively. Song and Xie (2009) reported similar results based on eight years of field observation in the same area (94.3% in citrus orchard and 84.3% in cropland). The subsurface lateral flow accounted for 35% to 42% of annual rainfall in the mono cropping system and for 14% to 34% in the agroforestry system, and the average was lower by 9.2% in the agroforestry system than in the mono cropping system. The reduced subsurface lateral flow was equivalent to the increase in precipitation interception in the agroforestry system, suggesting that the enlarged tree canopy was one of the reasons for the reduced subsurface lateral flow. In addition, the agroforestry system had a higher evapotranspiration than the mono cropping system; this can also reduce the generation of subsurface lateral flow.Estimating $NO_3$-N loss depended on the estimate of the subsurface lateral flow and the $NO_3$-N concentration measured in the field. Although there was no replication of the plots representing the land use systems, the results were reproduced in a temporal sense within the three monitoring years and showed larger subsurface lateral flow and higher $NO_3$-N concentration in the mono cropping system than in the agroforestry system. The annual $NO_3$-N loss through subsurface flow was estimated to be 4.9 times as much as the total nitrogen loss by soil erosion and overland flow (Table 5). The annual N loss ranged from 45 to 64 kg ha$^{-1}$ in the mono cropping system and from 16 to 48 kg ha$^{-1}$ in the agroforestry system. Similar results were reported in the literature. De Oliveira Leite (1985) demonstrated that subsurface lateral flow accounted for 14% of annual rainfall and led to an annual total nitrogen loss of 22.1 kg ha$^{-1}$ a$^{-1}$ on a cocoa cropping slope in Brazil. Honisch et al. (2002) reported that annual total nitrogen loss through subsurface lateral flow into the brook was estimated at 292–141 kg a$^{-1}$ from a 13-ha catchment with various arable land systems.Many studies have demonstrated the importance of controlling soil erosion (e.g. Palma et al., 2007b) and restoring riparian wetland (e.g. Larson et al., 2000) or forest (e.g. Schultz and Nicholas, 2004) to intercept and uptake eroded nutrients before they reach the streams. This study has highlighted the importance of controlling subsurface lateral flow, and the efficiency of agroforestry systems in reducing such flow as well as N loss. The agroforestry system studied here may have two mechanisms for reducing N loss, one controlling subsurface lateral flow and the other lowering the $NO_3$-N concentration due to lower input and greater uptake. In accordance with a similar assumption, agroforestry systems were proposed in southeast Australia to alleviate secondary salinity by intercepting subsurface lateral flow (Dunin, 2002). Further studies are needed to isolate the mechanisms why agroforestry system reduces $NO_3$-N loss through reduction of subsurface lateral flow and assess the environmental and agronomic effectiveness of agroforestry systems, particularly in the planning of proper landscape and land use (Palma et al., 2007a). To reduce non-point source pollution, it is imperative that one reduce N leaching, either through an optimization of N fertilization or by the introduction of deep-rooted trees to increase N uptake, and decrease subsurface lateral flow from the deep soil layers.

## 5. Conclusions

The spatial variations of soil water potential in rainstorm and during the three-year period demonstrated that the subsurface lateral flow generated along the hillslopes under both the agroforestry system consisting of peanut crop intercropped with citrus trees and the mono peanut cropping system in Jiangxi Province of China. Subsurface lateral flow generated as soil water was saturated over the low permeable soil layers and saturated soil water was retained to shallower soil layer for a longer time during a typical rainstorm in the agroforestry system than in the mono cropping system. Two-dimensional modelling reliably showed that the subsurface lateral flow accounted for 35% to 42% of annual rainfall in the mono cropping system and for 14% to 34% in the agroforestry system during the three year observation period, with an average reduction by 9.2% of annual rainfall in the agroforestry system. The reduced subsurface flow was equivalent to the amount of precipitation predicted to be reduced by interception, The two cropping systems received the similar amount of organic and inorganic N fertilizers (160.0–170.0 kg N ha$^{-1}$ a$^{-1}$), but the total amount of N inputs was lower in the agroforestry system as it may have received less biological fixed N by about 0 to 70 kg N ha$^{-1}$ a$^{-1}$ due to the smaller effective area occupied by peanut crops. $NO_3$-N concentration measured in soil water at all soil depths between 0.20 and 0.85 m ranged from 0.04 to 16.6 mg L$^{-1}$ in the agroforestry system and from 0.06 to 52.5 mg L$^{-1}$ in the mono cropping system. The mono cropping system had annually bimodal dynamics of $NO_3$-N concentration at all soil depths and higher $NO_3$-N concentration on the lower slope than on the upper slope position, suggesting vertical leaching and lateral movement of $NO_3$-N in subsurface soil due to excess N application. The scattered high $NO_3$-N concentration at 0.40 m depth and the decreased $NO_3$-N concentration at 0.85 cm depth in the agroforestry system indicated that $NO_3$-N leached to subsurface soil may be taken up by the deeper root, resulting in little accumulation of $NO_3$-N in the soil profile in the agroforestry system. The estimated $NO_3$-N loss associated with subsurface lateral flow ranged from 45 to 64 kg ha$^{-1}$ a$^{-1}$ in the mono cropping system and from 16 to 48 kg ha$^{-1}$ a$^{-1}$ in the agroforestry system. The smaller $NO_3$-N loss in the agroforestry system was probably attributed to the lower total amount of N inputs and/or to the reduced subsurface lateral flow. Lower N inputs will reduce $NO_3$-N leaching to deeper soil and the reduced subsurface flow will retain the leached $NO_3$-N in soil for a longer time during which it can be used by the deeper citrus' root system. The underlying mechanisms and the effectiveness of agroforestry systems at a large scale need further study. In general, the results in this study indicate that to reduce the environmental risk by $NO_3$-N pollution from slope lands with traditional mono cropping systems, either the fertilizer application rate, the time or the methods have to be improved, or the land use systems have to be changed from mono cropping systems with shallow roots to intercropping systems including deep-rooted trees.

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